



CANDIDATE
NAME

CG

INDEX NO

BIOLOGY

9744/03

Paper 3 Long Structured and Free-Response Questions

15 September 2021

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

Indicate the question you have attempted at the top of page 21.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, **submit booklets A, B and C separately** to the invigilator.

For Examiner's Use	
Section A	
1	27
2	13
3	10
Section B	
4 or 5	25
Total	75

This document consists of **25** printed pages and **3** blank pages.

Section A

Answer **all** the questions in this section.

- 1** Cystic fibrosis (CF) is one of the most common inherited genetic disease in humans. It is inherited as an autosomal recessive mutation.

CF is due to mutation of the cystic fibrosis transmembrane conductance regulator (CFTR) gene, which codes for CFTR protein. Most CF patients have a mutated CFTR gene on chromosome 7, consisting of deletion of three consecutive nucleotides, resulting in deletion of phenylalanine from CFTR protein.

CFTR protein functions as chloride ion channels embedded in membranes of cells that line the respiratory tract, pancreas and intestines. It regulates the transport of chloride ions out of these cells.

(a) With reference to the information given above,

(i) explain why cystic fibrosis (CF) is a genetic disease.

- 1. Illness which results from the mutation of a particular gene, CFTR gene and could be passed down to offspring;**
- 2. Loss of function mutation where both copies of the alleles must be mutated for diseases phenotype to be observed;**

[2]

(ii) describe the effect of the mutation of CFTR gene on the function of the CFTR protein.

- 1. Deletion of three consecutive nucleotides on the CFTR gene removes one codon from the mRNA,**
results in deletion of one amino acid (phenylalanine) from polypeptide sequence of CFTR protein, resulting in different bonds formed between amino acids thus different 3-dimensional configuration;
- 2. CFTR protein unable to transport chloride ions out;**

[2]

Almost 40% of adults with CF develop a form of diabetes known as cystic fibrosis-related diabetes (CFRD).

The bacterium *Pseudomonas aeruginosa* can cause chronic (long-lasting) lung infections. A person with CFRD is likely to have poorer lung function and a greater likelihood of having a chronic lung infection than a person who has CF but does not have CFRD.

An investigation was carried out to find out if the severity of damage to lung function in a person with CFRD is affected by

- their gender
- whether or not they have a chronic *P. aeruginosa* infection.

The investigators measured lung function by recording the maximum volume of air that can be expelled from the lungs in the first one second of a forced expiration. This is known as FEV₁. The lower the median FEV₁, the poorer the lung function.

Table 1.1 summarise the results of this investigation. All the 812 people in the study had cystic fibrosis.

Table 1.1

	Without chronic <i>P. aeruginosa</i> infection				With chronic <i>P. aeruginosa</i> infection			
	male		female		male		female	
	with CFRD	without CFRD	with CFRD	without CFRD	with CFRD	without CFRD	with CFRD	without CFRD
Number of people	44	110	52	93	106	166	121	120
FEV₁	71.1	71.4	53.6	73.6	49.0	59.0	42.0	61.0

(b) With reference to Table 1.1,

- (i)** discuss whether or not there appears to be a positive correlation between having a chronic *P.aeruginosa* infection and having CFRD.

1. yes, more people with infection have CFRD than those without infection;

2. [QV, no. of ppl] Without infection, 44 males / 52 females / 96 male & female have CFRD compared to with infection, 106 males / 121 females/ 227 male & female have CFRD;

3. [QV, FEV₁] Without infection, 71.1 in males / 53.6 in females / 124.7 in male & female compared to with infection, 49.0 males / 42.0 females/ 91.0 male & female;

4. [QV, % by gender] With no infection 28.5% males and 35.8% females have CFRD compared to with infection 38.9% males and 50.05% females have CFRD;

[2]

Point 1 compulsory, Point 2,3,4 either or;

- (ii) Calculate the percentage difference between the FEV₁ of males and females without CFRD and without *P.aeruginosa* infection.

Show your working.

$$73.6 - 71.4 / 71.4 \times 100; = 3.08\% / 3.1\% ;$$

OR

$$73.6 - 71.4 / 73.6 \times 100; = 2.99\% / 3.0\% ;$$

[2]

Fruit fly, *Drosophila*, has many different species.

Drosophila melanogaster are commonly used in labs for the studies of Type 2 diabetes progression.

Three of these species, *Drosophila pseudoobscura*, *D. persimilis* and *D. Miranda*, are thought to be closely related.

Samples of these three species were collected from the western United States of America. The base sequence of four regions of DNA of each species were sequenced. The divergence of these base sequences in *D. pseudoobscura* and *D. persimilis* from the sequences in *D. Miranda* was calculated.

The results are shown in Table 1.2.

Table 1.2

DNA region	<i>Drosophila</i> species	Percentage divergence of base sequence from that of <i>D. miranda</i>
1	<i>pseudoobscura</i>	2.5
	<i>persimilis</i>	2.4
2	<i>pseudoobscura</i>	8.1
	<i>persimilis</i>	7.3
3	<i>pseudoobscura</i>	2.1
	<i>persimilis</i>	1.7
4	<i>pseudoobscura</i>	1.9
	<i>persimilis</i>	1.7

- (c) With reference to Table 1.1, describe the evidence that *D. Miranda* may be more closely related to *D. persimilis* than to *D. pseudoobscura*.

1. divergence values less for *D. persimilis* than for *D. pseudoobscura* (at all DNA regions) or vice versa;

2. [QV] Any correct set of data for a particular DNA region;

[2]

- (d) Suggest why there is more divergence in some regions of DNA than in others.

1. there tends to be less divergence if DNA is part of an important gene or vice versa; / non-coding regions of DNA tend to accumulate more mutations than coding regions;

2. mutation in coding regions causes change in essential protein which will affect survival;

[2]

Plasma cells secrete antibody molecules in response to infections caused by bacterium such as *Pseudomonas aeruginosa*.

Fig. 1.1 is a diagram of an antibody molecule. The diagram is **not** complete.

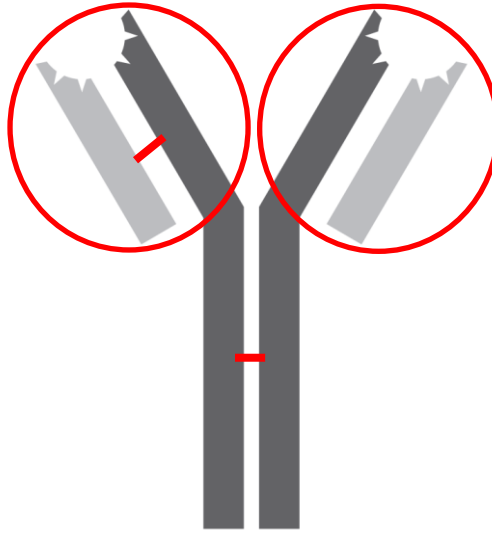


Fig. 1.1

(e) (i) Draw a circle around a variable region. [1]

(ii) Draw in and label the position of the disulfide bonds in the molecule. [1]

**Solid line(s) between one light polypeptide and one heavy polypeptide,
Solid line(s) between the two heavy polypeptides ;
maximum of six lines in each site**

(iii) Explain the importance of disulfide bonds in protein molecule, such as antibodies.

**1. (disulfide) bonds are between, cysteine(s) / cysteine residues ;
@ between R groups S-H S-H**

1. Covalent bond which is a strong bond that is not easily broken

2. Holds polypeptide(s)/ polypeptide chain(s)/ (protein) together in tertiary (quaternary) structure;

@ proteins/ strands

3. Maintain 3-dimensional configuration/ shape/ structure of protein molecule;

@ prevent deforming

@ structure unqualified

[3]

(iv) Describe how antibodies provide protection against bacterium.

1. Antibodies will bind with antigen on bacterium, at the variable region which contains antigen binding region specific to antigen;
2. Bacterium is neutralised, unable to attach to host cell receptor;
3. Opsonisation occurs where phagocytes will bind to Fc portion of antibodies to promote phagocytosis;
4. Natural killer cells are recruited to result in lysis of bacterium with the release of chemicals and proteins;
5. Agglutination/ aggregation of bacterium, prevents their ability to spread and invade tissues;

Point 1 is a compulsory point

[4]

- (f) DNA methylation plays an important role in development of disease and the process of aging. In a study, scientist examined DNA methylation at 476,366 sites throughout the genome of plasma cells from a population cohort (N = 421) ranging in age from 14 to 94 years old. Age affects DNA methylation at almost one third of the sites.

Fig. 1.2 shows the effect of DNA methylation with chronological age.

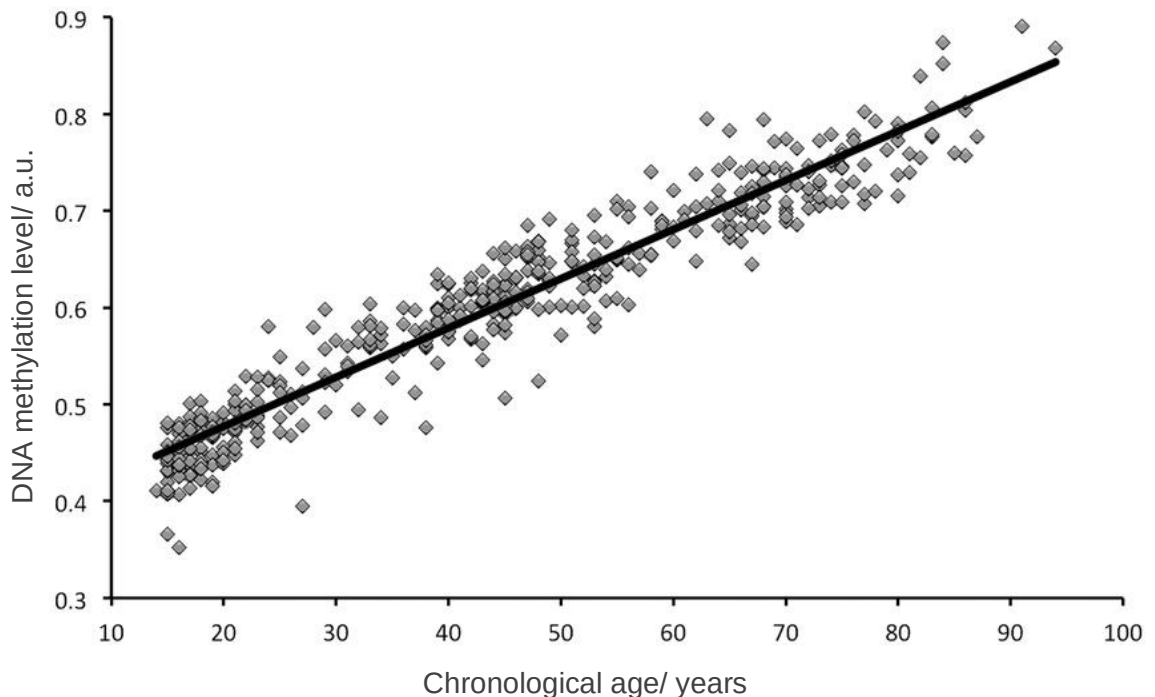


Fig. 1.2

- (i) With reference to the Fig. 1.2, discuss the relationship between aging, DNA methylation and immunity.

1. As age increases, DNA methylation level increased linearly as well;

2. QV correctly with units;

3. As age increases, increased DNA methylation indicates more gene silencing in plasma cells/ lesser genes in plasma cells are expressed;

4. Indicating that immunity decreases as we age;

[4]

- (ii) Briefly describe how can DNA methylation lead to cancer.

1. (Excessive) DNA methylation in tumour suppressor genes, leading to gene silencing;

Cell division is not inhibited in tumour cells,

2. thus allow to proliferate/ apoptosis of tumour cells are not triggered/ DNA repair not being carried out being cells divide;

[2]

[Total: 27]

- 2 In an experiment, genetic material from mouse liver tissues were isolated and treated with DNase 1. DNase 1 is an enzyme that can degrade double-stranded DNA in a non-specific manner. It was found that certain regions of DNA in the cell are naturally protected from degradation by DNase 1.

The DNase 1 enzyme was removed after digestion and any remaining intact DNA samples were extracted and placed under optimal conditions conducive for nucleic acid hybridization to occur.

Radioactively labelled probes complementary to certain genes were then added to the DNA samples. The amount of radioactivity in the samples were determined after excess probes were washed off.

- (a) With reference to the principles of nucleic acid hybridization, explain one step that must be taken first before hybridization of the probes can occur.

1. **expose intact DNA sample to high temperatures / high pH / alkaline solution,**
to denature the DNA / separate DNA strands by breaking hydrogen bonds
between complementary bases pairs;
2. **so that single-stranded DNA exposed**
so that the probes can complementary base pair with the exposed
bases;

[2]

The levels of hybridization of labelled probes to two specific gene sequences were measured via radioactivity in arbitrary units (a.u). Some of the results are shown in Table 2.1 below.

Table 2.1

Chromatin sample from liver cells	DNase I treatment	Gene complementary to radioactive probes	Radioactivity (a.u)
1	No	He1	89
2	Yes	He1	27
3	No	Av3	90
4	Yes	Av3	91

- (b) With reference to Table 2.1,

- (i) identify which gene, *He1* or *Av3*, is transcriptionally active in liver cells.

He1 is more transcriptionally active in liver cells;

[1]

(ii) explain your answer to (i).

1. Sample 2 shows the lowest radioactivity of 27au,

@ if students quote the difference in radioactivity between sample 1 and 2;

while sample 4 shows high radioactivity of 91au;

@ if students state radioactivity remained relatively unchanged at high levels

2. implying that He1 gene is found in regions less tightly packed / found in euchromatin,

thus making it more susceptible / accessible for DNase 1 to degrade;

3. while Av3 is found in regions more tightly packed / found in heterochromatin;

thus making it less susceptible / accessible for DNase 1 to degrade;

[3]

(c) The concentration and activity level of proteins can be regulated in different cell types through certain mechanisms.

Explain how regulation of either the concentration or activity level of proteins is carried out.

Concentration

1. For named chromatin / transcriptional / post-transcriptional control (DNA methylation / histone acetylation / presence of activators / increasing 3' poly A tail length)

2. Explanation of control; (change in chromatin structure resulting in ↑ or ↓ accessibility of RNA polymerase and transcription factors to polymerase leading to altered rate of transcription / bind to enhancers to up-regulate transcription / increases half-life of mRNA)

3. Effect of control (e.g. Increased transcription leads to more mRNA template for protein synthesis / more stable mRNA template allows more protein synthesis)

Or

4. Named translational control (e.g. repressors binding to 5' cap of mature mRNA);

5. Explanation of control (e.g. prevents binding of ribosomes to mRNA);

6. Effect of control (e.g. reducing translational rate, thus protein level decreases);

Or

7. Named post-translational control (Ubiquitination / adding of ubiquitin);

8. Explanation of control (degrade proteins that are worn out or defective);

9. Effect of control (Mark proteins for degradation by proteasome);

Or

Activity

10. Named post-translational control (eg. chemical modification / proteolytic cleavage);

11. Explanation of control (eg. synthesised protein modified with chemical group e.g. phosphate);

12. Effect of control (eg. non-functional protein turns functional);

[3]

Cancer is a disease in which normal controls over cell division are lost and malignant tumours form. An early diagnosis of many types of cancer can result in successful treatment.

The BRCA2 protein is involved in suppressing the development of tumours.

Several different dominant alleles of the gene, *BRCA2*, code for faulty versions of the BRCA2 protein. The presence of any one of these faulty alleles leads to an increased chance of developing several types of cancer, including liver cancer. Not everyone with one of these alleles develops cancer. This is because environmental factors, including lifestyle, are also involved.

Fig. 2.1 is a pedigree (family tree) showing the occurrence of cancers in four generations of a family. The presence of a faulty *BRCA2* allele was confirmed in person 15. The other individuals with cancer were not tested for the presence of the allele.

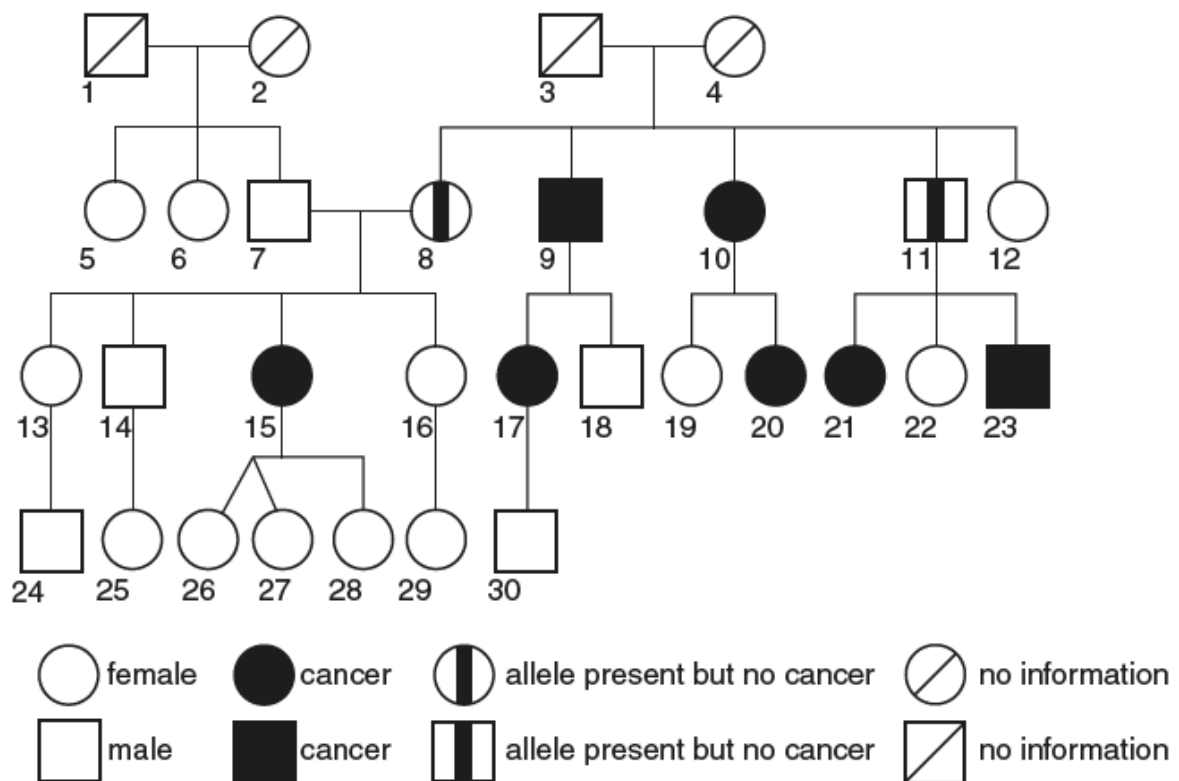


Fig. 2.1

- (d) Discuss the extent to which Fig. 2.1 provides evidence that a faulty *BRCA2* allele increases the risk of a person developing cancer.

Any 4 below:

-
1. individual 8 or 11 has *BRCA2* / allele, but does not have cancer ;

 2. no evidence / unknown, that individuals (apart from 15) with cancer have, *BRCA2* / allele
OR
individuals with cancer (apart from 15) may have a different mutation ;

 3. no children of individual 15, (known to) have the allele / have cancer ;

 4. individuals in fourth generation / children of individual 15, may develop cancer later in life ;

 5. individual 15 has cancer and, *BRCA2* / allele ;

 6. some) individuals with cancer in third generation had a parent with cancer
OR
(some) individuals with cancer in third generation had a parent with, *BRCA2* / allele ;

 7. individual 3 or 4 may have had the *BRCA2* / allele
OR
any individual from 8 to 11 may have inherited, *BRCA2* / allele, from 3 or 4 ;

 8. idea that overall data is inconclusive; [4]

[Total: 13]

- 3 Tuberculosis (TB) is an infectious disease caused by a bacterial pathogen.

Fig. 3.1 shows the infection cycle of the pathogen.

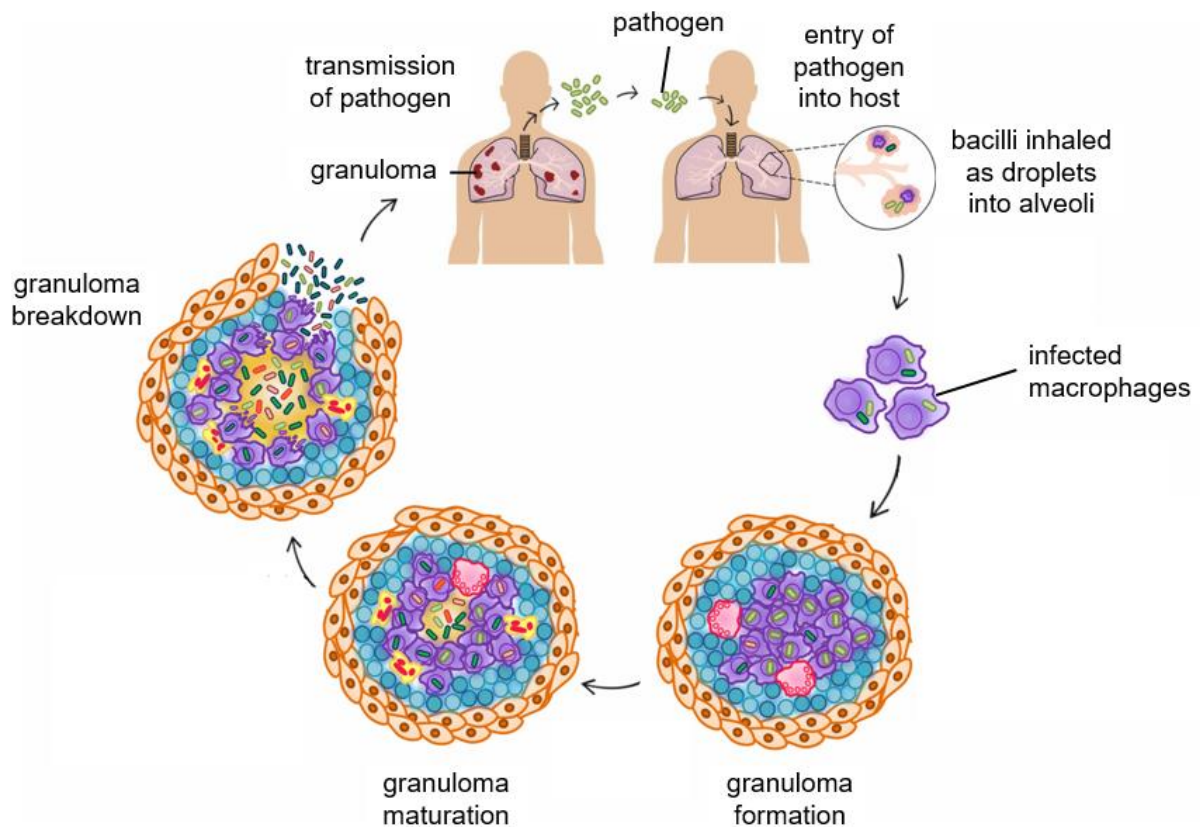


Fig. 3.1

- (a) (i) Name the pathogen that causes tuberculosis.
Mycobacterium tuberculosis ; (underline) [1]
- (ii) With reference to Fig. 3.1, describe how latent TB is established in an individual upon infection.
1. alveolar macrophages phagocytose bacteria
 (inhibit fusion of phagosome with lysosome) evade killing and continue to survive within macrophage ;
 2. release cytokines ➡ attract neutrophils and macrophages
 granuloma formation ➡ bacteria persist in dormant stage ; [2]

The global increase in drug-resistant tuberculosis has prompted a search for alternative therapies.

Researchers hypothesised that the use of a combination of antibiotics is useful in treatment of drug-resistant tuberculosis. In the investigation, researchers identified the minimum inhibitory concentration (MIC) of antibiotics against multiple strains of the pathogen. The antibiotics used in each experimental set-up are as follows:

- **A:** amoxicillin only
- **B:** a combination of amoxicillin and clavulanate (amoxicillin/clavulanate)
- **C:** meropenem only
- **D:** a combination of meropenem and clavulanate (meropenem/clavulanate)

Amoxicillin and meropenem are β -lactum antibiotics. Clavulanate is a β -lactamase inhibitor, inhibiting the action of β -lactamase that is secreted by antibiotic-resistant bacteria.

The MIC refers to the lowest concentration of antibiotic being tested that inhibits the growth of bacteria. A higher MIC indicates that the bacteria are resistant against the antibiotic, while a lower MIC indicates that the bacteria are more susceptible towards the antibiotic.

Fig. 3.2 shows the results of the investigation.

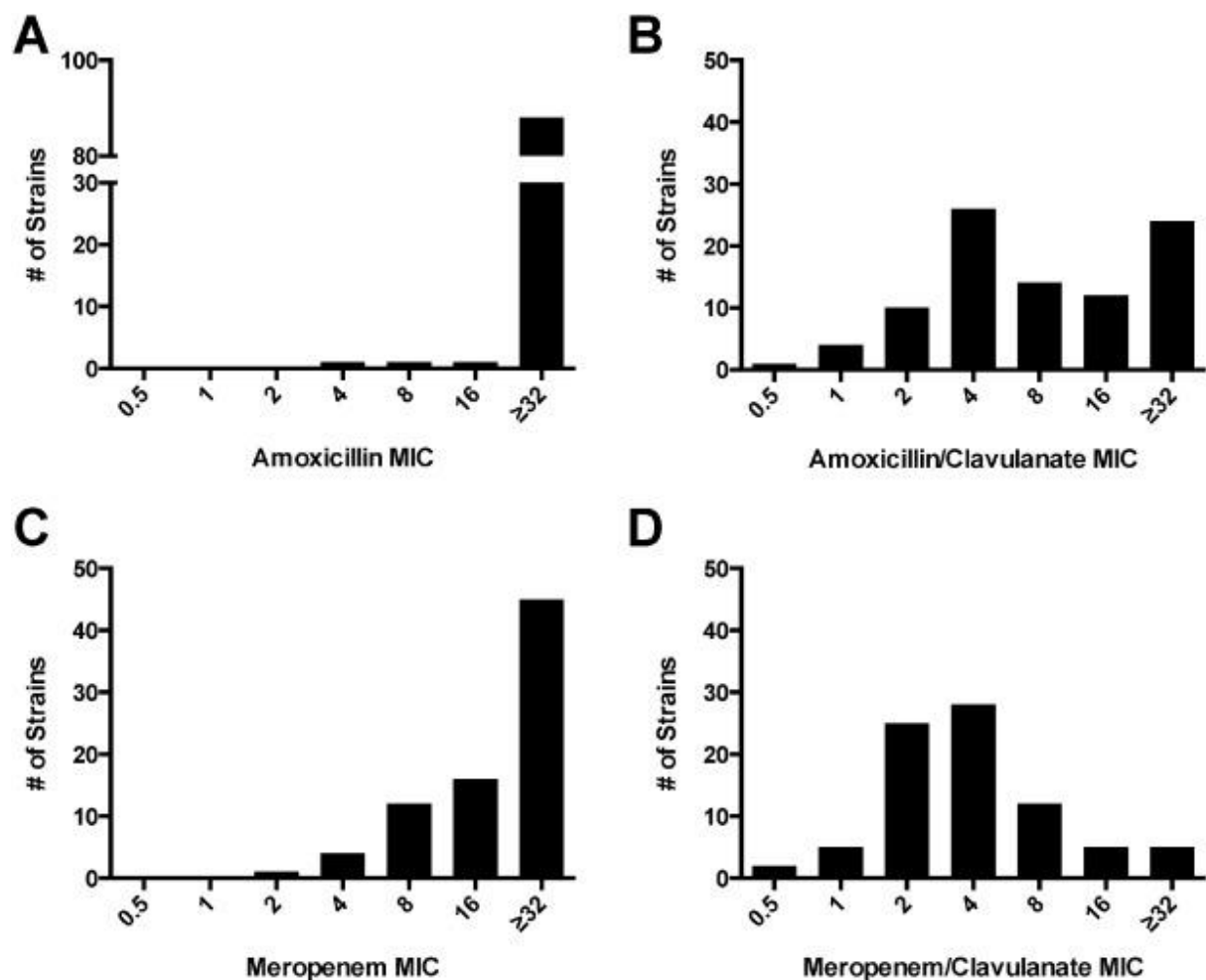


Fig. 3.2

- (d) With reference to Fig. 3.2 and the information provided, discuss whether the results of the investigation support the researchers' hypothesis.

1. supports hypothesis

as amoxicillin/clavulanate and meropenem/clavulanate have more strains with a lower MIC $\therefore$ bacteria more susceptible to combination of antibiotics ;

2. [QV] e.g. 10 strains treated with amoxicillin/clavulanate and 25 strains treated with meropenem/clavulanate have MIC of 2,
vs 90 strains treated with amoxicillin only and 45 strains treated with meropenem only have a higher MIC of >32 ;

3. antibiotic-resistant bacteria secrete β -lactamase against the β -lactam antibiotics

$\therefore$ clavulanate inhibits β -lactamase secreted $\rightarrow$ allow β -lactam antibiotics to kill / inhibit growth of bacteria / OWTTE;

[3]

- (e) Penicillin, like amoxicillin and meropenem, is a β -lactam antibiotic.

Describe the mode of action of penicillin on bacteria.

1. binds to active site of transpeptidase

prevent enzyme from cross-linking peptidoglycan chains ;

2. resulting in weak cell wall $\rightarrow$ collapse and disintegrate when the cell divides

killing bacteria via osmotic lysis ;

[2]

- (f) Peptidoglycan and cellulose are important components of cell walls in bacteria and plants respectively.

Compare the structures of peptidoglycan and cellulose.

feature	peptidoglycan	cellulose
1. monomer	NAM (N-acetylmuramic acid), NAG (N-acetylglucosamine)	β -glucose ;
2. crosslinks	short peptides	H bond cross-links ;
3. chains	both are made of polysaccharide chain that are cross-linked with other chains ;	
4. type of biomolecule	both are polysaccharides / made of monosaccharides joined by glycosidic bonds ;	

® transpeptidase (catalyse peptide cross links) vs cellulose synthase (catalyse polymerisation of β -glucose)

[2]

[Total: 10]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

You answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

- 4 (a) Explain the various roles of membranes in cells. [12]

1. Membranes are made up phospholipid bilayer with hydrophobic core;
2. Protein embedded within phospholipid bilayer;
3. Presence of unsaturated fatty acids in hydrocarbon core of membranes, and cholesterol in hydrocarbon core of animal membranes;

Feature	Roles
Selectively permeability	4. Being selectively permeable, membrane only allows some molecules to cross directly without assistance; 5. Non-polar/uncharged molecules are able to dissolve and diffuse through; 6. Ions/charged ions and/or most polar molecules are repelled by hydrophobic core of membrane;
Membrane fluidity	7. Maintained by presence of unsaturated fatty acids in plant / bacterial membranes (and) also cholesterol in animal membranes; 8. Enables formation of transport vesicles for transport of proteins formed at bound ribosomes on rER to Golgi apparatus / sER for further modification; 9. Enables formation of transport / secretory vesicles for transport of glycoproteins / secretory products from Golgi apparatus to cell surface membrane; 10. Facilitates fusion of membranes of vesicles and organelle membranes / cell surface membranes for embedment of proteins in membranes / for exocytosis;
Increased surface area	11. Highly folded inner mitochondrial membrane increases surface area to hold more proteins required for oxidative phosphorylation; 12. e.g.: electron transport chains/ ATP synthase in cristae of mitochondria;
Feature	Roles

Compartmentalisation	13. Formation of unique environment for highly specialised activities; 14. e.g. acidic environment maintained in lysosomes enabling enzymes to function; 15. localisation of enzymes for reactions to take place so that they are not supposed to catalyse reactions that they are not supposed to; 16. e.g.: enzymes in Calvin Cycle in stroma; enzymes in Krebs cycle in mitochondrial matrix; 17. accumulation of ions to generate a concentration gradient; 18. e.g. high H^+ concentration in intermembrane space in mitochondria/within thylakoid lumen in chloroplasts versus lower H^+ concentration in mitochondrial matrix / stroma of chloroplast establishes a proton gradient for chemiosmosis and formation of ATP;
proteins of related function arranged together in membrane	19. Allows proteins of related function to be grouped together to enhance sequential biochemical processes; 20. e.g. electron carriers in electron transport chain (ETC) in inner mitochondrial membrane are arranged in order of increasing electronegativity / decreasing energy levels to facilitate flow of electrons down ETC during oxidative phosphorylation;
Cell-cell recognition and adhesion	21. Glycoproteins / glycolipids are involved in cell-cell recognition; 22. (any 1 e.g.) e.g.: T cell receptor on membranes of naïve T cells + how it helps to recognize peptide: MHC complex on antigen presenting cells;
Signal transduction	23. Some transmembrane proteins serve as cell surface receptors to transfer information from environment into cell when specific molecules (ligands) bind to them; 24. E.g. glucagon (hormone) bind to glucagon receptor in membrane which triggers a cascade of chemical reactions leading to hydrolysis of glycogen to glucose; E.g. insulin
QWC: Ref. to 3 different features	

- (b) Discuss how antibody diversity is generated in B cells, and how plasma cells are able to secrete specific antibodies against a particular antigen. [13]

(A) haematopoietic stem cells give rise to naïve B-cells (max 3m)

1. (in bone marrow) haematopoietic stem cells differentiate to give rise to lymphoid progenitor cells ;
2. somatic recombination occurs, resulting in combinatorial diversity ;
3. gene rearrangement to produce different combinations of VDJ and VJ gene segments for heavy chain and light chain respectively ;
4. imprecise joining resulting in gain or loss of nucleotides ➔ junctional diversity ;
5. ref to test for self-reactivity ;
6. produce wide variety of naïve B cells with specific B cell receptors / membrane-bound immunoglobulin (Ig) ;

(B) activation of naïve B-cell (max 4m)

7. antigen that is complementary to antigen-binding site (on variable region) of specific B cell receptor able to recognise and bind ;
8. B cell present antigenic peptide on MHC to helper T cells with TCR complementary to antigenic peptide ➔ activation of B cells ;
9. ref to affinity maturation ;
10. somatic hypermutation occurs ➔ introduce mutation to rearranged Ig genes ;
11. B cells expressing Ig with higher affinity to Ag selected to proliferate (while those expressing Ig with lower affinity to Ag undergo apoptosis) ;
12. ref to class switching ;
13. gene rearrangement resulting in change in constant region of heavy chain ;
14. clonal expansion and differentiation of B cells to form plasma cells and memory cells ;

(C) secretion of Ab by plasma cells (max 4m)

15. transcription of heavy chain gene and light chain gene to produce mRNA ;
16. translation of mRNA by bound ribosomes on rER membrane to produce polypeptide ;
17. 2 identical heavy chains and 2 identical light chains assemble to form functional immunoglobulins;
18. packaged into transport vesicle which buds out from rER membrane and fuses with cis face of GA ;
19. chemical modification of Ig ;
20. modified Ig packaged into secretory vesicle which buds out from trans face of GA ;
21. transported to CSM with aid of cytoskeleton, fuses with CSM and releases Ig via exocytosis ;

QWC: at least 2m from each section

- 5 (a)** Ions are an atom or molecule with a net electric charge.

Discuss the role of ions in ensuring the continuity of life.

[13]

(A) role of H^+ ions in photosynthesis / respiration

1. contributes to accumulation of H^+ in thylakoid lumen in chloroplast / intermembrane space in mitochondria ;
2. generate proton gradient ➡ generate proton motive force ;
3. H^+ diffuse back into stroma / mitochondrial matrix via ATP synthase, phosphorylating ADP to ATP via chemiosmosis ;
4. combines with (final electron acceptor) O_2 and electrons to form water during oxidative phosphorylation ;
5. combines with electrons with coenzymes NADP / NAD / FAD to form reduced coenzymes ;

(B) role of ions in DNA

6. phosphate group in DNA is negatively-charged ;
7. allow formation of ionic bonds with positively-charged histones (to form nucleosome core) ;

(C) role of ions in proteins

8. amino acids exist as zwitterions in solution @ diagram of zwitterion ;
9. in acidic solution, able to accept H^+ to form NH_3^+ ;
10. in basic solution, able to donate H^+ to form COO^- ;
11. R groups of amino acids can be acidic / negatively charged or basic / positively charged ;
12. allow formation of ionic bonds in tertiary / quaternary structure ;

(D) other ions

13. ref to Fe^{2+} in Hb ➡ allow binding to O_2 for transport ;
14. ref to Ca^{2+} act as second messengers in cell signalling pathways ;
15. ref to Mg^{2+} act as cofactors in enzymatic reactions ;
16. ref to Mg^{2+} in chlorophyll ➡ allow absorption of light ;
17. AVP (e.g. Na^+ and K^+ ions transported by sodium-potassium pump to maintain ion concentration for nervous response) ;

QWC: at least 2m in 2 different sections ;

- (b)** Using *lac* operon as an example, explain how gene regulation is achieved in prokaryotes. [12]

1. lac operon is a cluster of structural genes coding for enzymes involved in lactose metabolism ;
2. *lacZ* codes for β -galactosidase, *lacY* codes for permease, *lacA* codes for transacetylase ;
3. regulated by a single operator and promoter ;
4. ref to inducible operon ;

(A) negative regulation by lactose

5. in high concentration of lactose, lactose is converted to allolactose by β -galactosidase ;
 6. allolactose acts as inducer, binds to lac repressor causing it to become inactive ;
 7. lac repressor detaches from operator ;
 8. RNA pol able to bind promoter to transcribe structural genes ;
- @ reverse argument

(B) positive regulation by glucose

9. in high concentration of glucose, adenylyl cyclase is inhibited ➡ low concentration of cAMP ;
 10. cAMP binds to CAP to form active cAMP-CAP complex ;
 11. cAMP-CAP complex binds to CAP-binding site in promoter ;
 12. facilitate binding of RNA pol to promoter ➡ increase transcription of structural genes ;
- @ reverse argument

QWC: at least 2m from (A) and 2m from (B) ;